



Smart Home Automation Roadmap

Complete Documentation & Execution Plan



Project Overview

Name: Smart Home Automation using ESP32

Goal: Build a low-cost, scalable system to control household appliances remotely through IoT platforms and future AI integration.



System Architecture

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graph TD; A[Phone / Web / Voice] --> B[Internet (Blynk Cloud)]; B --> C[ESP32 Wi-Fi Controller]; C --> D[Relay Board -> Appliances];
```



Hardware & Software

Category	Component	Purpose
Hardware	ESP32 Dev Board	Main Brain / Wi-Fi Connectivity
Hardware	4-Channel Relay	Physical Switching of Loads

Category	Component	Purpose
Hardware	5V Adapter	Power Supply
Software	Arduino IDE	Firmware Development
Software	Blynk IoT	App & Cloud Dashboard
Software	Google Assistant	Voice Control Integration



Wiring Connections

ESP32 Pin	Relay Pin
VIN / 5V	VCC
GND	GND
GPIO 23	IN 1
GPIO 22	IN 2
GPIO 21	IN 3
GPIO 19	IN 4



Development Phases



Phase 1 & 2: Planning & Assembly

- Define appliance count and relay capacity.
- Assemble hardware: Connect ESP32 to Relay and test with low-voltage bulbs.

✓ Phase 3 & 4: Coding & App Setup

- Configure Blynk Template ID and Auth Token.
- Define Virtual Pins (V0: Light, V1: Fan, V2: Socket, V3: Aux).
- Upload ESP32 firmware via Arduino IDE.

✓ Phase 5 & 6: Integration

- Set up Blynk Web Dashboard for PC control.
- Link Google Home/Assistant for voice commands.

🔒 SAFETY GUIDELINES

- Always test circuit with 5V/12V DC before connecting AC 110V/220V.
- Ensure all AC wiring is properly insulated with heat shrink or electrical tape.
- Do not touch the relay board or ESP32 while the project is plugged into AC mains.

🚀 Long-Term Vision

The system is designed to be the foundation for a **Personal JARVIS Smart Room**. Future upgrades include motion sensing for auto-lights, temperature monitoring for AC control, and custom LLM integration for smart conversational home control.